A large EEG dataset with ECG, skin conductance, respiration, gaze and pupillometry for emotion decoding

Authors: NeuroAI team

Abstract

We present a multimodal emotion dataset comprising 383 unique participants, 345 recording hours and 252 video stimuli. An image-based questionnaire guided stimulus selection according to each participant's emotional associations. Synchronized measurements include 64-channel electroencephalography (EEG), electrocardiography (ECG), skin conductance, respiration, gaze, and pupil size. Annotations include reported emotional categories, valence and arousal ratings, and dominance ratings for a subset of the data. The dataset supports research on emotional processing, individual differences, and the development and evaluation of emotion-decoding models.

Introduction

Emotion decoding aims to infer aspects of a person's emotional state from measurable signals, such as brain activity, heart activity, or breathing. This research can help us understand how emotions are expressed in the brain and body and develop technologies that respond to users' emotional states. Machine learning offers tools for finding patterns in these signals, but training and evaluating such models requires *labels*: reference descriptions of the emotional responses associated with each recording.

Obtaining dependable labels is a central challenge. The same video may evoke amusement in one person and discomfort in another. Researchers can collect participants' reports of their experiences, but these reports do not provide direct access to the experiences themselves. They depend on participants' ability to recognize, remember, interpret, and describe what they felt. Emotional responses also involve bodily changes influenced by the sympathetic and parasympathetic branches of the autonomic nervous system. Physiological signals provide more objectively recorded measurements, although interpreting their relationship to emotion remains challenging. Both approaches offer evidence about emotional responses, with different strengths and limitations.

Public datasets such as FACED, SEED-IV, and SEED-VII have supported emotion decoding using electroencephalography (EEG), which records electrical activity at the scalp, with both SEED datasets additionally providing eye-tracking and pupil-size measurements. However, emotion categories assigned to videos may differ from what individual viewers experience. When recordings associated with the same videos appear in both training and test data, models can also recognize patterns specific to those videos. High classification accuracy may consequently reflect stimulus recognition without establishing emotion decoding. Our analysis of FACED demonstrated this problem, illustrating the importance of stimulus diversity and annotations that account for individual responses (Gerster et al., 2026).

The dataset comprises 383 participants and 690 recording sessions. Before recording, participants associated stimulus themes with emotions using an image-based questionnaire, and corresponding videos formed personalized stimulus sets. Session numbers and video counts varied across cohorts. After each video, participants rated valence and arousal and, where collected, dominance, and could report up to three emotions from 12 categories.

Synchronized recordings combine 64-channel EEG with ECG, skin conductance, respiration, gaze, and pupil size. Together with questionnaire information, these data support studies of emotional responses and individual differences, as well as the development and evaluation of affective decoding models.

Methods

Participants and ethics

The final dataset comprises 383 unique participants, corresponding to 690 recordings when repeated participation was considered. Of these, 271 were female, and the mean age 39.6 ± 12 years (Table 1). Participants were recruited from an external recruitment agency's panel and received financial compensation for their participation, administered by the agency.

To be eligible for inclusion, participants were required to be at least 18 years old, report good physical and mental health, be proficient in French, and have the legal capacity to provide informed consent. Participation in one cohort did not preclude participation in another. Consequently, a single individual could be assigned multiple participant IDs.

Exclusion criteria included pregnancy or breastfeeding, known allergies or immunocompromising conditions, and diabetes, where applicable to the specific experimental protocol. Participants with hearing impairments or color vision deficiencies were excluded when these conditions could interfere with the experimental tasks. Participants were required to have hairstyles that permitted proper EEG cap fitting and adequate electrode–scalp contact. Scalp scars or other conditions that interfered with electrode impedance were exclusion criteria. Participants with recently dyed hair were required to have washed their hair at least three times following dye application to minimize dye transfer and staining of the EEG equipment. Participants were instructed to arrive with clean hair, free of styling products, and without facial makeup, facial oils, strong perfumes, or strongly scented deodorants. They were also required to abstain from smoking, energy drinks, coffee, alcohol, and recreational or psychoactive substances for at least 3 hours before completing the online psychological profiling questionnaire, and at least 12 hours before the experimental session. Written informed consent was obtained from all participants.

Table 1. Participant demographics in the final dataset

Characteristic	**Value**
Participants, *n*	N=383
Female, *n*	271
Age, years, mean ± SD	39.6 ± 12

Table 2. Cohort division of the final dataset

	Cohort 1	**Cohort 2**	**Cohort 3**	**Cohort 4**	**Cohort 5**
Video presentation order	Fully randomized	Fully randomized	Randomized emotion blocks	Randomized emotion blocks	Randomized emotion blocks
Valenca/Arousal/Dominance Rating	Whole-numbers	Continuous-scale	Continuous-scale	Continuous-scale	Continuous-scale
Number of sessions (days)	2	2	3	2	1

	Cohort 1	Cohort 2	Cohort 3	Cohort 4	Cohort 5
Dominance rating	No	No	Yes	Yes	Yes

Recording setting and protocol

Recordings took place in the laboratory at the Human Augmented Brain System office in Paris, France. Participants completed one to three sessions depending on the cohort (Table 2), with repeat sessions typically separated by approximately 7 days. Sessions lasted approximately 60 minutes and included emotional-video trials and other tasks. The released data comprise selected task segments: video viewing, resting state, and, where available, an artefact calibration block. Segment availability and duration varied across sessions.

EEG was recorded using a 64-channel Waveguard™ Net (*NA-265.10; ANT Neuro*) with electrodes arranged according to the extended international 10–10 layout. The system used Ag/AgCl ring electrodes with saline-soaked sponge interfaces with shielded cables. Signals were acquired at 1,000 Hz with an eego™ 64 amplifier (*EE-225; ANT Neuro bv, Hengelo, the Netherlands*).

The online reference electrodes were placed at Cz and CPz, while ground electrodes were placed at AFz and Fpz.

Electrode impedances were checked using eego™ software (*version 01.11.04; ANT Neuro)*, with an impedance threshold set at 60 kΩ. To ensure good electrode–skin contact and minimize impedance, participants were asked to comb their hair before electrode placement. The scalp was cleaned with alcohol wipes to prepare the electrode sites, and a small amount of conductive gel was applied to the ground and reference electrodes to improve electrical contact. Hair was carefully parted at each electrode site to ensure direct contact between the saline-soaked sponges and the scalp.

Additional physiological signals, including electrocardiography (ECG), respiratory effort, and galvanic skin response (GSR), were acquired at a sampling rate of 1,000 Hz using a SenseBox unit (XS-271; ANT Neuro) connected to the amplifier. ECG was recorded using two Ag/AgCl disposable surface electrodes (*KendallTM H124SG; Cardinal Health*) placed bilaterally in the infraclavicular region. Respiratory activity was monitored using a piezoelectric effort band (*1370-kit; SleepSense*) positioned around the lower thorax, just above the abdomen. EDA was recorded using two electrodes attached to the middle phalanges of the index and middle fingers of the non-dominant hand.

Eye tracking and pupil size were measured using Pupil Labs Neon glasses at an approximate sampling rate of 200 Hz. Eye-tracking and pupillometry data were subsequently upsampled to 1,000 Hz for integration with the other modalities. The released multimodal BDF files contain EEG, additional physiological and eye-tracking channels, and an event status channel; channel mappings and units are provided in the accompanying record maps.

Multimodal data acquisition and synchronization were managed using Lab Streaming Layer (LSL) to facilitate the temporal alignment of data streams. All data streams were simultaneously recorded using LabRecorder and stored in XDF format, preserving timestamps and synchronization information for subsequent processing.

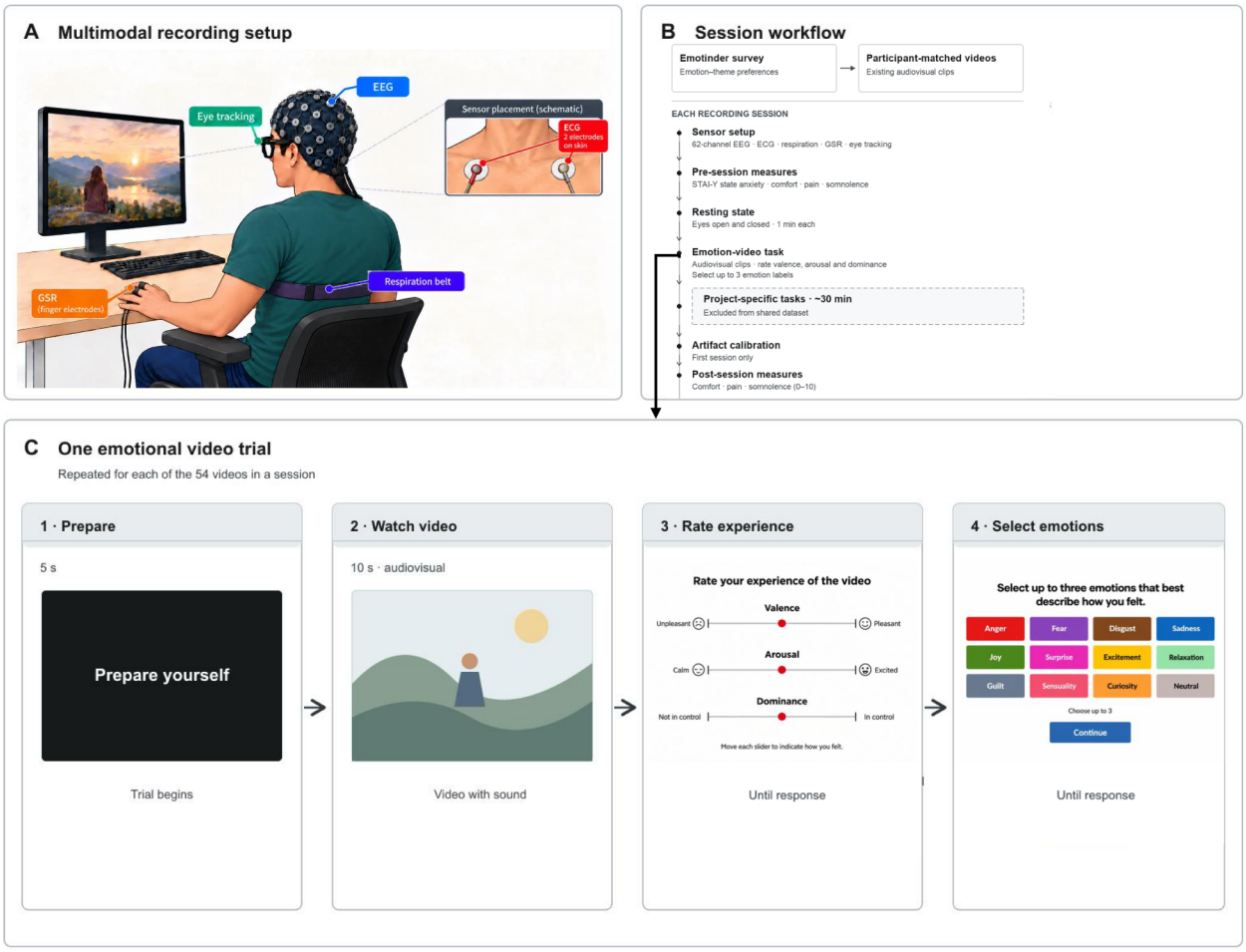


Figure 1. Recording setup and participant procedure. (A) Multimodal signals were recorded while participants viewed audiovisual stimuli. (B) Session workflow, including resting state and, where available, artefact calibration. (C) Video trials included preparation, video viewing, ratings, and selection of up to three ranked emotion labels. Rating availability varied by cohort.

Preprocessing

The Multimodal recordings were preprocessed in Python using our internal library. Recordings were cropped to the interval between the first and last event markers, retaining up to 5 s of additional data at each boundary where available. EEG channels T9 and T10 were excluded before filtering and independent component analysis (ICA).

A 50 Hz notch filter with a quality factor of 30 was applied to ECG, respiration, and skin conductance signals to attenuate power-line interference. Gaze and pupillometry signals were excluded from this filtering step. EEG signals were subsequently bandpass filtered from 0.5 to 48 Hz using a zero-phase finite impulse response filter following the EEGLAB-style FIR filtering approach, then rereferenced to the common average of the retained EEG channels.

ICA was performed on the continuous, filtered, and rereferenced EEG using the Picard algorithm through EEGPrep. Electrode coordinates were supplied from the recording montage, and the number of independent components was set to the estimated rank of the EEG data. Picard was run in extended, non-orthogonal mode with whitening and centering enabled, a maximum of 500 iterations, a convergence tolerance of 10^{-7}, and a fixed random seed of 5489.

Independent components were classified using ICLabel. Components were rejected when their estimated probability was at least 0.75 for eye artefacts or at least 0.90 for muscle artefacts, line noise, or channel noise. Components meeting any of these criteria were removed by subtracting their reconstructed contributions from the EEG signals. The ICA decomposition, component classifications, and rejected component indices were saved alongside text and JSON reports documenting component removal. Finally, the cleaned EEG was reintegrated into the synchronized multimodal recording and exported in BDF format.

Emotional stimuli and Individualized Stimulus Selection

The initial stimulus design comprised 252 ten-second videos across 84 themes and 12 target emotional categories: anger, fear, disgust, sadness, joy, surprise, excitement, relaxation, guilt, sensuality, curiosity, and neutral. Each theme was associated with a predefined default emotion and initially contained three representative videos (Table 3).

Although standardized affective stimulus databases, such as the International Affective Picture System (IAPS), the Open Affective Standardized Image Set (OASIS), and the Nencki Affective Picture System (NAPS), provide validated emotional stimuli, they do not necessarily account for interindividual variability in emotional associations. Indeed, stimuli that typically evoke a given emotion at the population level may elicit different, or even opposing, emotional responses across individuals. To account for this variability and increase the likelihood of eliciting the intended emotional responses, an individualized stimulus selection procedure was implemented.

Prior to the experimental recording sessions, participants were presented with representative images corresponding to each of the video themes. For each image, participants were asked to identify the emotion they most strongly associated with the depicted theme from among the 12 predefined emotional categories. Participants were subsequently asked to select, for each emotional category, the three themes they considered most representative of that emotion.

Based on these individual selections, a personalized video set was generated for each participant. For each of the 11 non-neutral emotional categories, nine videos were selected, corresponding to the three videos associated with each of the three chosen themes. In contrast, the neutral category consisted of nine predefined videos that were identical for all participants, ensuring a standardized neutral condition across the dataset.

The standard selection procedure targeted 108 videos per participant (99 personalized emotional videos and nine standardized neutral videos). Depending on the experimental protocol, videos were distributed across one, two, or three recording sessions, and the number of videos varied (Table 2).

When a participant selected fewer than three themes for a given emotional category, the remaining themes were selected from the predefined theme–emotion associations (Table 3), ensuring that no theme was assigned to more than one emotional category for the same participant.

The video presentation order was randomized within the protocol used for each cohort (Table 2). For protocols involving multiple sessions, selected videos were distributed across sessions.

Table 3. Default themes per emotion category

Emotion	Themes
Anger	Hate symbols; Animal suffering; Police brutality; Violence; Discrimination; War; Traffic jams; Accidents; Vandalism
Fear	Fear of heights; Horror; Clowns; Negative jump scares; Confined spaces; Snakes; Spiders; Negative jump scares
Disgust	Vomiting; Wounds; Toilets; Teeth; Insects; Carcasses; Moldy food; Waste; Sweating; Diseases; Surgery
Sadness	Sad people; Old age; Cemeteries; Disasters; Dead plants; Exploitation; Animal suffering; Injustice; Broken objects; Abandoned places
Joy	Animals; Happy people; Couples; Children; Celebrations; Beautiful landscapes; Party decorations; Plants; Graduation
Surprise	Explosions; Positive jump scares; Thunderstorms; Costumes; Nudity; Eating
Excitement	Victory; Concerts; Extreme sports; Elections; Positive jump scares; Yoga; Reading
Relaxation	Cozy environments; Peaceful environments; Spa; Relaxing interiors; Gardens; Injustice; Waste; Breathing exercises
Guilt	Exploitation; Bullying; Broken objects; Homelessness; Finger-pointing; Flowers; Female flirting; Accidents; Eating
Sensuality	Nudity; Male flirting; Tenderness; Perfumes; Silk; Unusual landscapes; Unusual animals
Curiosity	Optical illusions; Unusual objects; Mechanisms; Art; Mysterious phenomena; Crowds staring; Optical illusions
Neutral	Table; Chair; Cutlery

Participant ratings and label processing

Participant-reported emotional responses

Following video presentation, participants rated valence and arousal on scales ranging from 1 to 9 and, where collected, dominance. Valence represented pleasantness or unpleasantness, arousal reflected perceived activation, and dominance reflected perceived control. Rating format and the availability of dominance varied by cohort (Table 2). Participants could also select up to three emotions from the 12 predefined categories, ranked by perceived relevance; the availability of these responses varied across trials and cohorts.

Each trial was associated with available ratings (rating_valence, rating_arousal, and rating_dominance), ranked emotional selections (self_reported_emotion_1 to self_reported_emotion_3), the predefined stimulus category (default_emotion_label), and the participant-specific expected category (expected_emotion_label).

Label refinement and quality screening

A rule-based procedure combined the default stimulus emotion, the participant-specific expected emotion, up to three ranked self-reported emotions, and estimates derived from the distribution of valence and arousal ratings. The resulting final_emotion_label is a derived annotation and may differ from any one input. Intermediate density-derived scores were used during refinement but are not separate columns in *labels.csv*. The detailed method and decision tree are provided in Supplementary Material.

Potentially atypical or inconsistent annotations were identified using label agreement, repetitive rating patterns, and valence–arousal outlier checks. Their combined indicator, flagged_outliers, marks trials for review; it does not establish that a response is incorrect or inattentive. A zero does not guarantee that all underlying ratings were available. Further details and thresholds are provided in Supplementary Material.

Technical Validation

The distribution of valence and arousal ratings, first-ranked emotion reports, and video-level ratings was examined to describe annotation coverage and class balance. The corresponding analyses and Figures 4–6 are provided in Supplementary Material.

Data Records

File organization

The dataset is organized into two main directories: *CURATED*, containing unprocessed recordings, and *PREPROCESSED*, containing cleaned and preprocessed data. The dataset root also contains loading scripts, a README and three CSV files with trial labels, marker definitions and participant information.

Within *CURATED*, each participant has a separate directory. Files are identified by participant, recording day and task. Each task recording consists of a multimodal .bdf file and a matching .record_map file, which is used by the supplied loading code to interpret the channels. A metadata .json file is provided for each available day.

The task segments are emotion-video (video viewing), resting-state and artefact (artefact calibration). Available days and segments vary between participants. The video files themselves are not provided.

File pattern	Contents
<participant>_day-<day>_metadata.json	Session event metadata and segment export information.
<participant>_day-<day>_task-<task>_multimodal.bdf	Multimodal time-series data for an available task segment.
<participant>_day-<day>_task-<task>_multimodal.record_map	Channel and modality mapping associated with the matching BDF.
labels.csv	One row per video trial with markers, ratings, and emotion annotations.
markers.csv	Lookup of marker codes and associated tasks or emotion categories.

File pattern	Contents
participants.csv	One row per participant ID with demographics and questionnaire information.

Session metadata and timing

The metadata JSON contains an events list and a dataset_export section. In the supplied export, events.onset_s uses the canonical full-session time reference. To obtain the onset relative to a segment, subtract that segment's metadata_time_origin_s: segment onset = events.onset_s − metadata_time_origin_s. For example, 1533.051 − 369.510 = 1163.541 seconds within the emotion-video segment. Check that an event falls within the segment before extracting it.

Metadata field	**Description**
events: marker_id, onset_s, duration_s	Event marker, onset in canonical session seconds, and duration in seconds.
onset_source, duration_source, trial_type_source	Provenance of event timing or type; values depend on the export.
trial_type, media_file, instruction_text	Event type, video filename when applicable, and instruction or event text.
emotion_label, rating_tool, valence_score, arousal_score	Annotations associated with the event; may be null.
dataset_export: participant_id, protocol, day, source	Export identifiers and provenance.
segments: file, record_map	Paired segment BDF and record-map filenames.
metadata_time_origin_s, source_time_origin_s, source_to_metadata_offset_s	Segment time origins and offset used for alignment.
requested_start_s, requested_end_s, source_start_s, source_end_s	Requested and source segment boundaries.
duration_s, n_samples, alignment_max_residual_s	Segment duration, sample count, and alignment residual in seconds.
event_indices, partial_event_indices	Event-index lists; availability and indexing require confirmation.

Trial-level label fields

labels.csv contains one row per video trial. The same video may occur more than once for a participant. Join to the relevant session using the standardized participant ID, day, and event marker after verifying the final export. Empty cells indicate unavailable values, not zero. A final emotion label is a derived annotation and need not equal the first-ranked self-report.

Field	Definition
cohort	Numeric identifier of the acquisition cohort or study grouping.
participant_id	Participant identifier, standardized across release files
gender	Recorded participant gender; missing when not available.
day	Recording-session number; missing for some trials in the current annotation export.
video	Filename or identifier of the presented video stimulus.

Field	Definition
start_marker	Event marker code associated with video onset; verify against session metadata and marker lookup.
end_marker	Event marker code associated with video offset; verify against the marker lookup.
final_emotion_label	Derived emotion label from the multi-source refinement procedure.
rating_valence	Participant-reported valence rating (scale: 1–9).
rating_arousal	Participant-reported arousal rating (scale: 1–9).
rating_dominance	Participant-reported dominance rating (scale: 1–9).
self_reported_emotion_1	First-ranked emotion selected by the participant after the video.
self_reported_emotion_2	Second-ranked emotion selected by the participant. Optional.
self_reported_emotion_3	Third-ranked emotion selected by the participant after the video. Optional.
default_emotion_label	Predefined emotion associated with the video stimulus.
expected_emotion_label	Emotion expected for that participant and video, based on personalized calibration or default assignment.
expected_emotion_source	Source of the expected label: “emotinder” for personalized selection or default for fallback mapping.
video_theme	English-language theme or category assigned to the video.
video_presentation_order	Presentation position where available; ordering conventions may vary across cohorts.
flagged_outliers	Combined binary screening indicator; 1 marks a trial for review and 0 does not establish that all inputs were available.

Marker lookup

markers.csv is a lookup table for marker codes; it does not provide participant-specific event times. Use the session metadata for event timing and verify the correspondence with the recording status channel.

Field	Description
Emotion or task	Task instruction or emotion category associated with the marker pair.
Start Marker	Marker code for the start of the task or trial.
End Marker	Marker code for the corresponding end.
Repetition	Repetition index in the marker mapping.

Participant-level metadata

participants.csv contains one row per participant ID. Demographic and questionnaire responses and scores are included where available; coverage varies by cohort and session. The participant identifier links these data to the other files once final keys are standardized.

Fields	Description
cohort, participant_id	Study cohort and participant identifier.
Demographic and other question-text fields	Age, gender, education and other demographic responses.

Supplementary Material

Detailed label processing

Identification of potentially non-attentive rating behavior

Constant repetition of the same rating was perceived as potential non-attentive rating behavior. To identify potentially repetitive rating behavior, two indicators were calculated separately for each participant and recording session, after rounding valence and arousal ratings to the nearest integer.

The first indicator was the proportion of trials within a session for which rounded valence and arousal ratings were identical. The second indicator was the proportion of trials assigned the most frequently occurring rounded valence–arousal pair within that session.

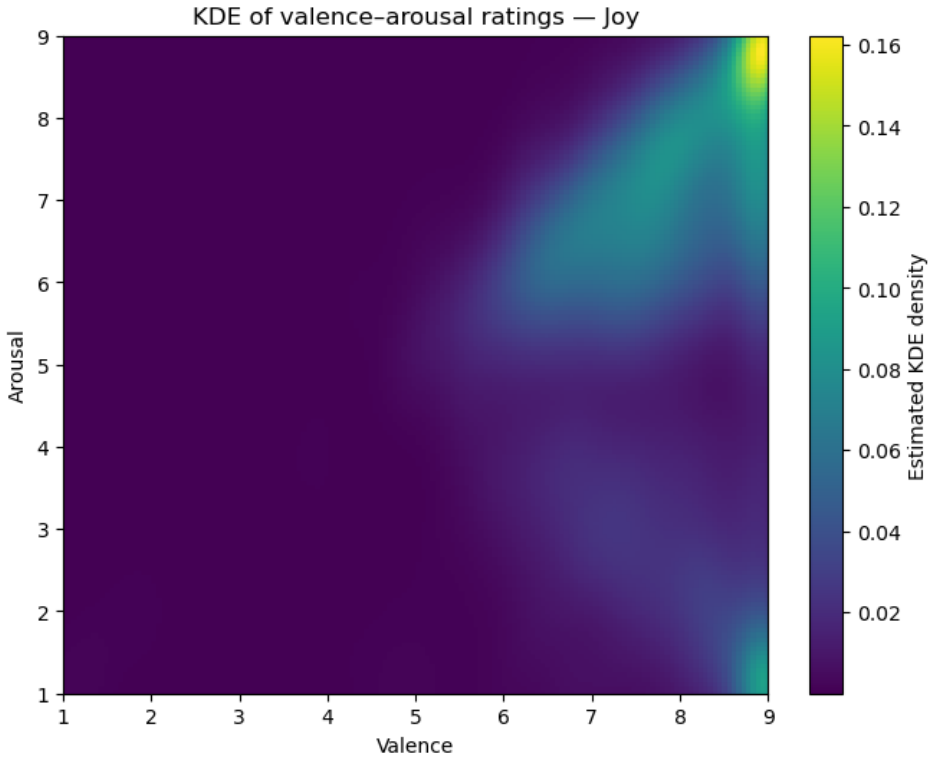
For each indicator, the mean and standard deviation of the session-level proportions were calculated across participant-session observations. Each session's proportion was then standardized relative to this distribution. Sessions exceeding the corresponding mean by more than 1.5 standard deviations were flagged as exhibiting unusually repetitive response patterns, and marked as positive *flagged_outliers*.

The threshold of 1.5 standard deviations was selected empirically following inspection of the distribution of session-level repetition rates. These indicators were used to identify potentially atypical rating behavior rather than to establish inattention definitively, as repeated ratings may also reflect genuine emotional experiences.

Estimation of emotion-specific valence–arousal distributions

Two-dimensional Gaussian kernel density estimations (KDEs) were fitted separately for each of the 12 emotions, using valence–arousal ratings of trials on which that emotion was selected first (self_reported_emotion_1). SciPy *gaussian_kde* was used with a bandwidth scaling factor of 0.2.

Supplementary Figure 2 illustrates the estimated valence–arousal density for trials in which joy was selected as the first-ranked emotion. The same procedure was applied independently to each emotion category.



Supplementary Figure 2. Kernel density estimation of valence–arousal ratings associated with joy. The heatmap represents the two-dimensional Gaussian KDE fitted to valence and arousal ratings from trials in which participants selected joy as their first-ranked emotion. The x- and y-axes represent valence and arousal ratings, respectively, on scales ranging from 1 to 9. Color intensity indicates the estimated probability density, with higher values corresponding to regions containing a greater concentration of ratings. Equivalent KDEs were constructed for each of the 12 emotional categories and subsequently used to derive relative emotion-membership scores for individual trials.

For each trial, its valence–arousal coordinates, denoted by M = (v, a), were evaluated against each emotion-specific density function (*f*). The resulting density values were normalized across the 12 emotional categories to obtain relative KDE-derived emotion-membership scores.

For an emotion *e*, the normalized membership score was defined as:

$$M_e(v,a) = \frac{f_e(v,a)}{\sum_k f_k(v,a) + \varepsilon}, \quad \varepsilon = 10^{-12}$$

Where $f_e(v,a)$ is the KDE density evaluated at the trial's valence–arousal coordinates for emotion *e*.

The emotion with the highest membership score was designated as the KDE-predicted emotion. The normalized scores were also ranked to identify the smallest set of emotions whose cumulative membership reached at least 78%.

The KDE-derived membership scores, KDE-predicted emotion, and cumulative membership set were intermediate variables used during emotion-label refinement. They are not included as separate columns in the released dataset.

Multi-source emotion label refinement

A rule-based procedure combined four sources of information to determine each trial's final emotion label: the predefined stimulus emotion (default_emotion_label), the participant-specific expected emotion (expected_emotion_label), up to three ranked self-reported emotions (self_reported_emotion_1, self_reported_emotion_2, and self_reported_emotion_3), and the intermediate KDE-derived estimates.

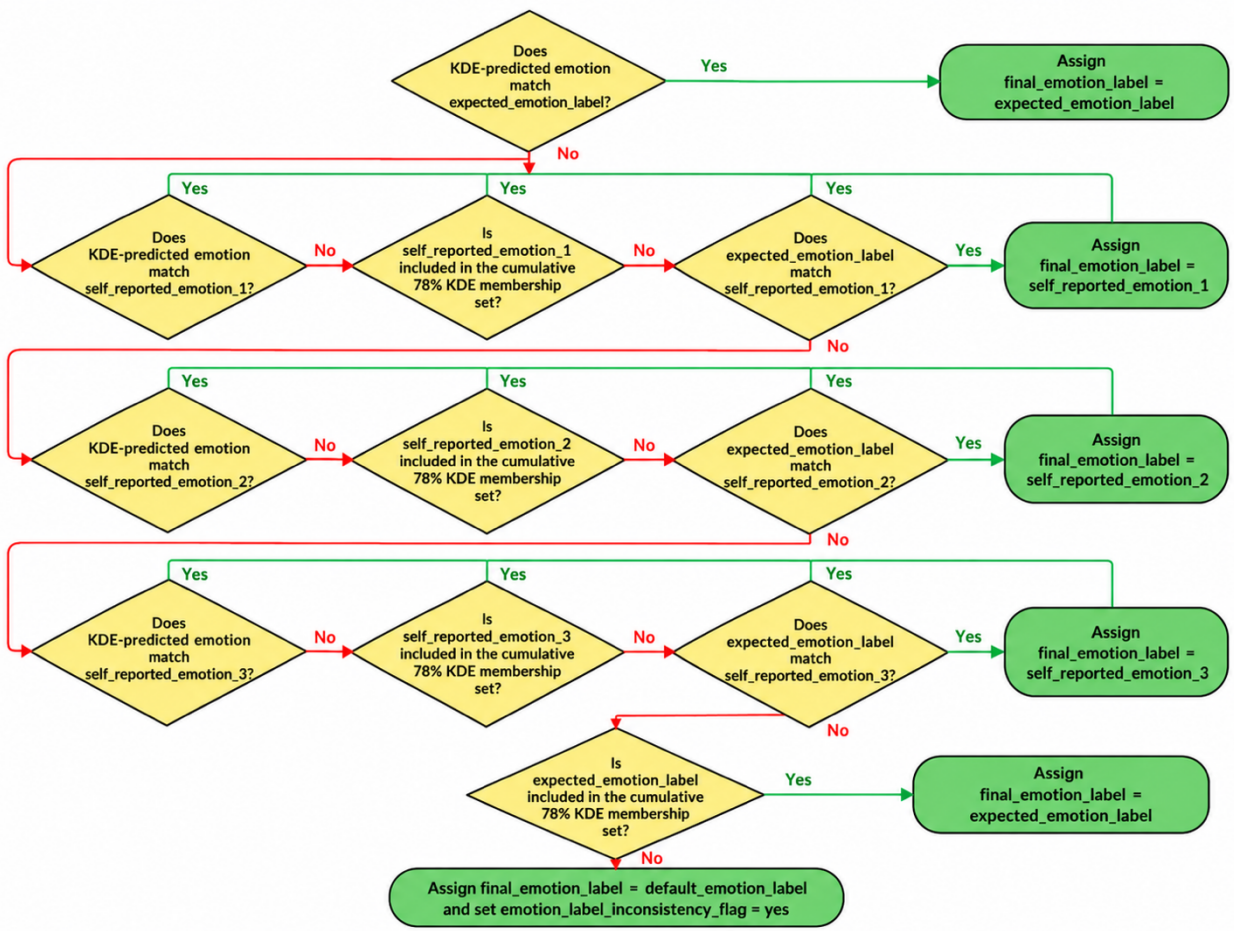
The algorithm first tested whether the KDE-predicted emotion matched the participant-specific expected emotion. If the two agreed, their shared emotion was directly assigned as the final emotion label (final_emotion_label).

Otherwise, the participant's self-reported emotions were evaluated sequentially, starting with the first-ranked response. A self-reported emotion was retained if it satisfied at least one of three conditions: (1) it matched the KDE-predicted emotion; (2) it belonged to the cumulative 78% KDE membership set; or (3) it matched the participant-specific expected emotion. Once a self-reported emotion satisfied one of these conditions, it was assigned as the final label, and no further self-reported emotions were evaluated.

If none of the three self-reported emotions satisfied these conditions, the participant-specific expected emotion was selected when it belonged to the cumulative 78% KDE membership set. Otherwise, the predefined stimulus emotion was retained and the trial was flagged as potentially inconsistent and and marked as positive *flagged_outliers*.

When valid KDE estimates were unavailable, a separate fallback procedure used the available emotion annotations.

The resulting label was stored in the released dataset as final_emotion_label. Supplementary Figure 3 summarizes the principal decision rules. Only the final label is provided in the released dataset; intermediate KDE-derived scores and predictions are not retained as separate variables.



Supplementary Figure 3. Rule-based decision tree for multi-source emotion label refinement. When valid KDE membership scores are available, the algorithm first checks whether the KDE-predicted emotion matches the participant-specific expected emotion. If they agree, that emotion is directly assigned as the refined label. Otherwise, the algorithm sequentially evaluates the three ranked self-reported emotions against the KDE prediction, the cumulative 78% membership set, and the expected emotion. If none satisfies the criteria, the expected emotion is reconsidered before reverting to the predefined stimulus emotion and flagging the trial as potentially inconsistent. A separate fallback procedure is applied when valid KDE scores are unavailable.

Identification of atypical valence–arousal responses

Following emotion-label refinement, Mahalanobis distances were calculated in the two-dimensional valence–arousal space to identify atypical ratings relative to two reference distributions. First, trials were grouped according to their final emotion label (final_emotion_label). Second, trials were grouped by the combination of video and final emotion label.

For each group containing at least five valid valence–arousal observations, the group mean and covariance matrix were estimated. Trials with Mahalanobis distances exceeding the empirical 97th percentile of distances within their respective reference groups were marked as positiive *flagged_outliers*. Groups with fewer than five valid observations were not evaluated.

These criteria identified observations that deviated from the corresponding reference distributions without implying that such responses were necessarily incorrect.

Descriptive analyses of annotations

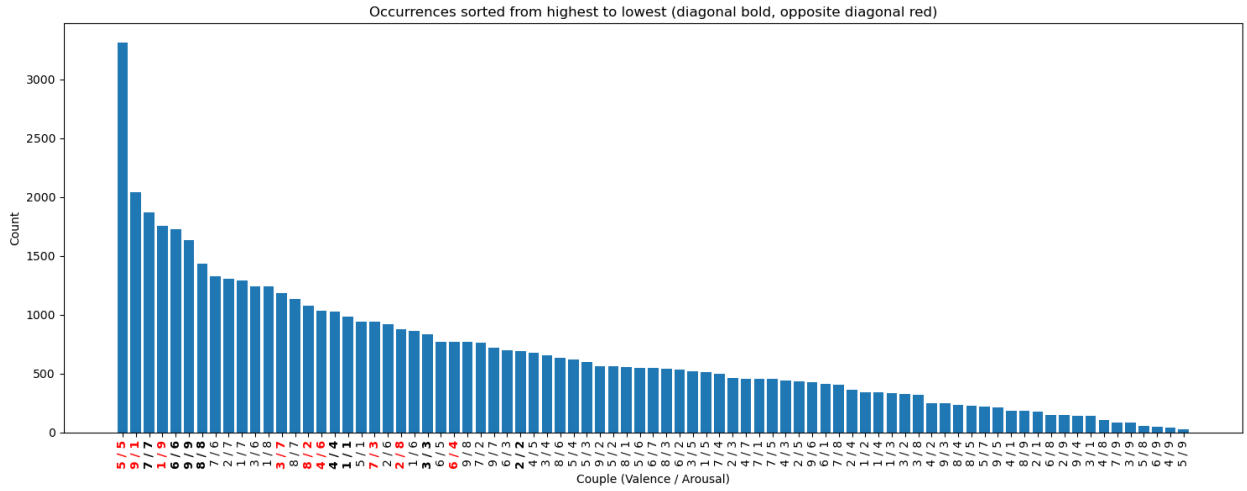
Distribution of valence–arousal ratings

To characterize the distribution of affective ratings across the dataset, we first examined the frequency of valence–arousal combinations reported by participants. Valence and arousal were both assessed on a nine-point scale, ranging from 1 to 9. Supplementary Figure 4a presents the frequency of rounded valence–arousal pairs, ordered from most to least frequent. The distribution is heterogeneous, with some combinations occurring substantially more frequently than others. In particular, several combinations along the negative valence–high arousal and positive valence–high arousal regions are frequently represented, alongside ratings near the midpoint of both scales.

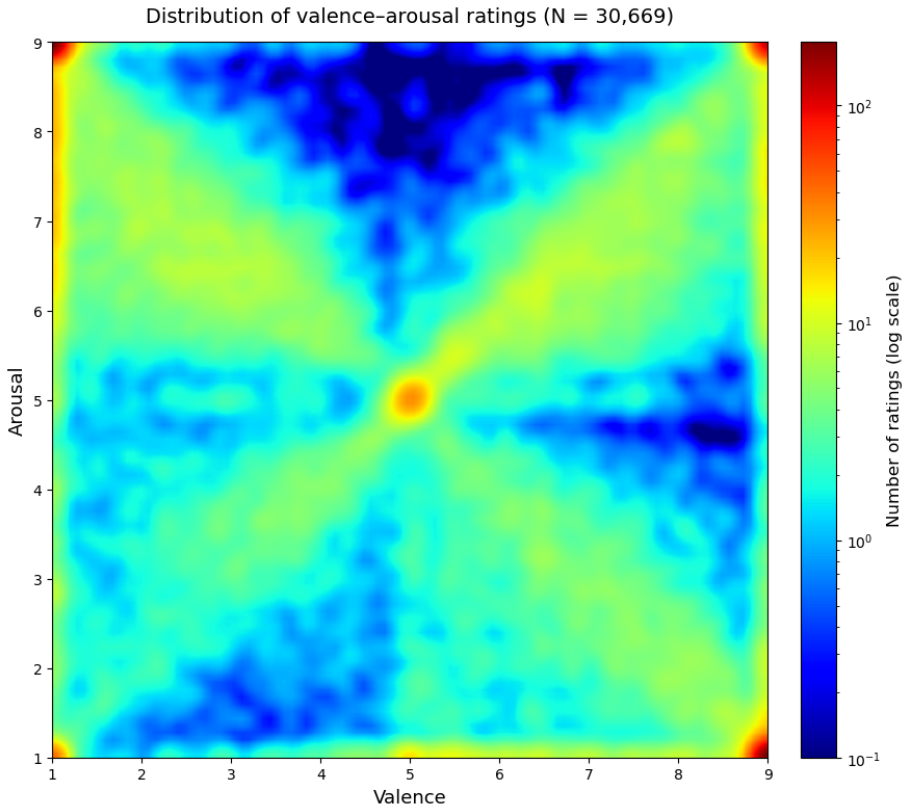
To further characterize the coverage of the affective space, we generated a two-dimensional heatmap of valence–arousal ratings (Supplementary Fig. 4b). The heatmap shows that observations are distributed across the full rating space, although their density is not uniform. Ratings tend to concentrate in particular regions, including the central area around valence and arousal values of 5, and extend toward the lower- and higher-valence extremes. The representation also reveals variations in rating density across the arousal dimension.

Together, these visualizations provide an overview of the range and relative frequency of subjective affective responses represented in the dataset.

a.



b.



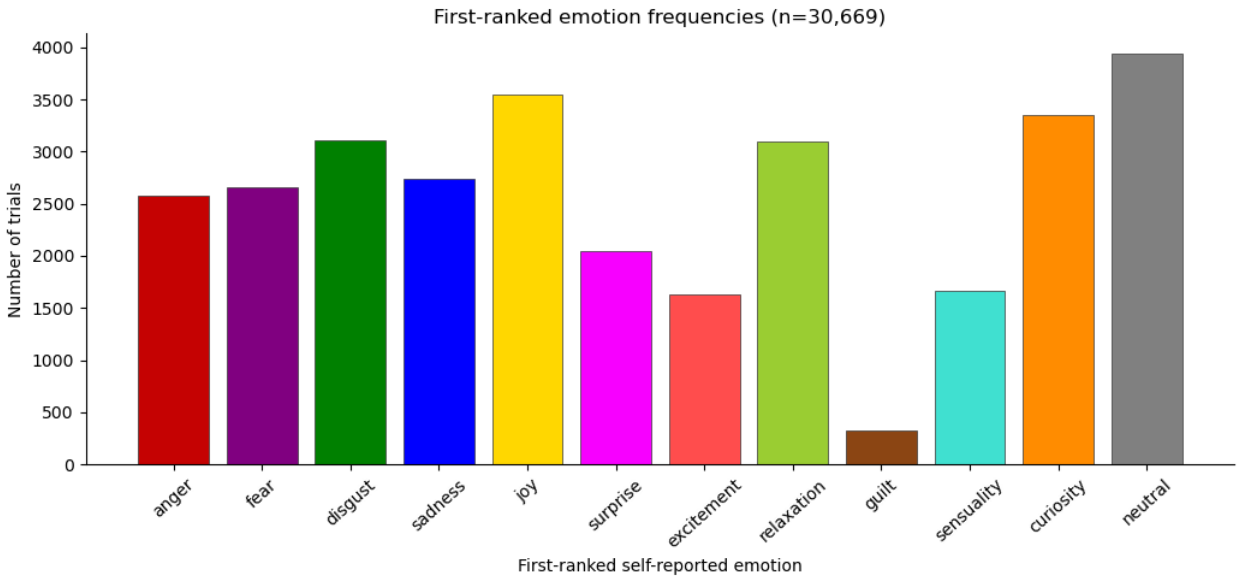
Supplementary Figure 4. Distribution of participant-reported valence and arousal ratings. (a) Frequency of rounded valence–arousal rating combinations, ordered from highest to lowest occurrence. Combinations with equal rounded valence and arousal are highlighted in bold, whereas combinations with opposite ratings are highlighted in red. **(b)** Two-dimensional heatmap of valence–arousal ratings (n = 30,669 valid observations, excluding cohort 1 and flagged outliers), displayed using a logarithmic color scale. Warmer colors indicate regions with higher concentrations of ratings, whereas cooler colors indicate less frequently represented regions. Both dimensions range from 1 to 9.

Distribution of first-ranked self-reported emotions

Following each video presentation, participants were asked to identify up to three emotions describing their subjective experience, ranked in order of relevance. To characterize the distribution of these annotations, we examined the frequency of emotions selected in the first-ranked position (Supplementary Fig. 5).

All 12 emotion categories were represented among the first-ranked responses, although their frequencies varied considerably. Neutral was the most frequently selected category, followed by joy and curiosity. Disgust, relaxation, sadness, fear, and anger also accounted for substantial proportions of responses, whereas surprise, excitement, sensuality, and particularly guilt were less frequently selected.

This distribution illustrates the variability in participants' subjective emotional responses and highlights an imbalance in the frequency of emotion categories. Such differences are relevant when interpreting emotion-specific analyses and developing affective decoding models, as categories with fewer observations may provide less information for model training and evaluation.



Supplementary Figure 5. Distribution of first-ranked self-reported emotions.
Number of trials assigned to each of the 12 emotion categories based on participants' first-ranked post-video responses (n = 30,669), flagged outliers excluded. Bars are colored according to emotion category. The distribution reveals differences in the representation of individual emotions, with neutral, joy, and curiosity occurring most frequently and guilt occurring least frequently.

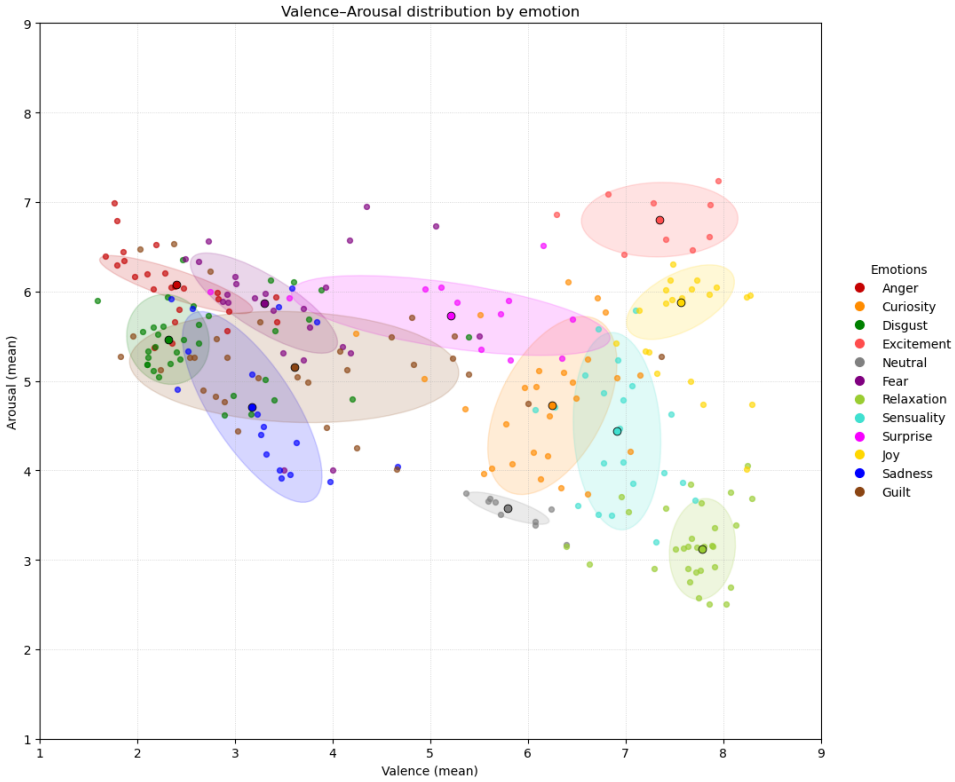
Valence–arousal distributions associated with default emotion labels

To evaluate the correspondence between predefined stimulus labels and participant-reported affective responses, we examined the distribution of valence and arousal ratings across videos grouped by their default emotion labels.

For each video, mean valence and arousal values were calculated by averaging the ratings provided by participants who viewed that video. Because the selection of stimuli was personalized, the number of ratings contributing to each video-level mean varied.

Supplementary Figure 6 presents the video-level means in the two-dimensional valence–arousal space, grouped according to the 12 default emotion categories. Robust covariance ellipses were estimated separately for each emotion using the Minimum Covariance Determinant (MCD) method, with a nominal coverage of 68%. The ellipses provide a descriptive representation of the location, dispersion, and orientation of the video-level rating distributions associated with each default label.

The distributions show that videos associated with different default emotion labels occupy partly distinct regions of the valence–arousal space. For example, videos labeled as joy and excitement tend to receive relatively high arousal ratings, while joy is primarily associated with positive valence and anger with negative valence. Relaxation is predominantly associated with positive valence and lower arousal, whereas sadness is generally located in the negative-valence region with moderate to lower arousal. Other categories exhibit varying degrees of overlap, reflecting the fact that distinct emotional experiences may share similar valence–arousal characteristics.

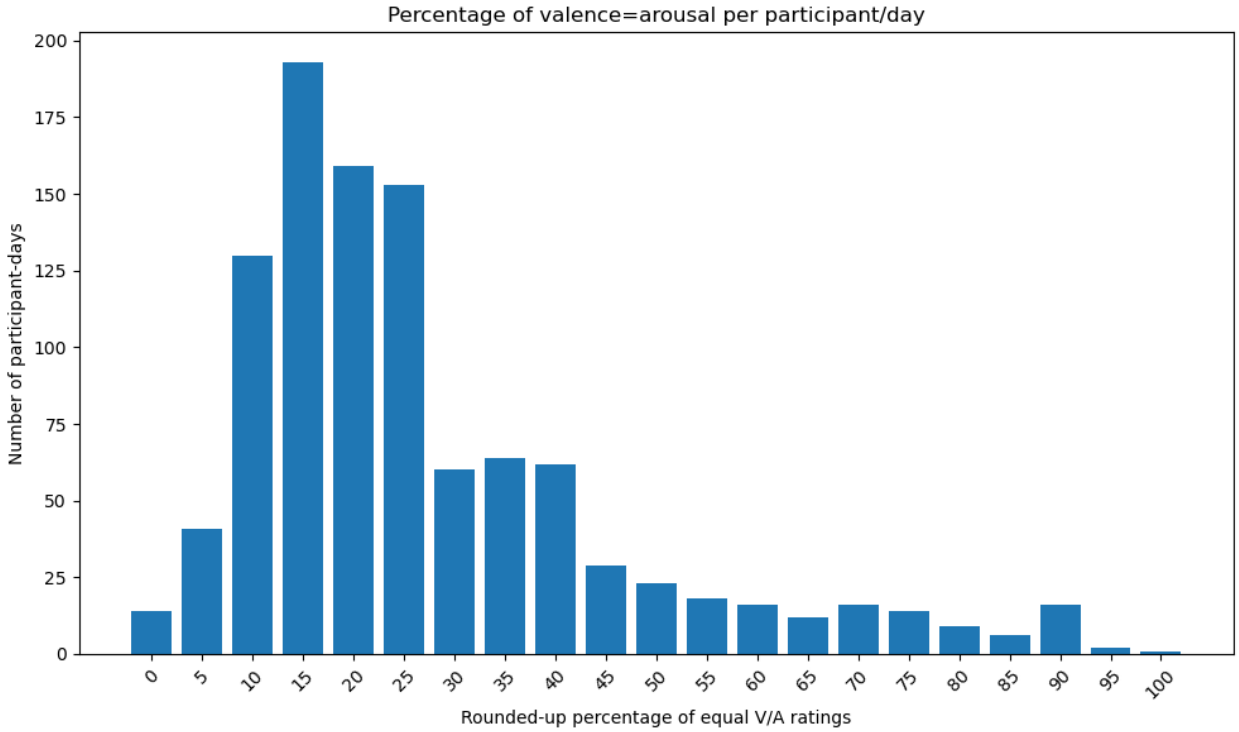


Supplementary Figure 6. Distribution of mean valence and arousal ratings by default emotion label. Each point represents a video, positioned according to its mean participant-rated valence and arousal and colored according to its predefined emotion category. Shaded ellipses represent robust covariance regions estimated using the Minimum Covariance Determinant method, with 68% nominal coverage. Colored central markers indicate the estimated robust centroids of the corresponding video-level distributions. Valence and arousal range from 1 to 9. The number of participant ratings contributing to each video-level mean varies. (252 videos, 54 579 ratings).

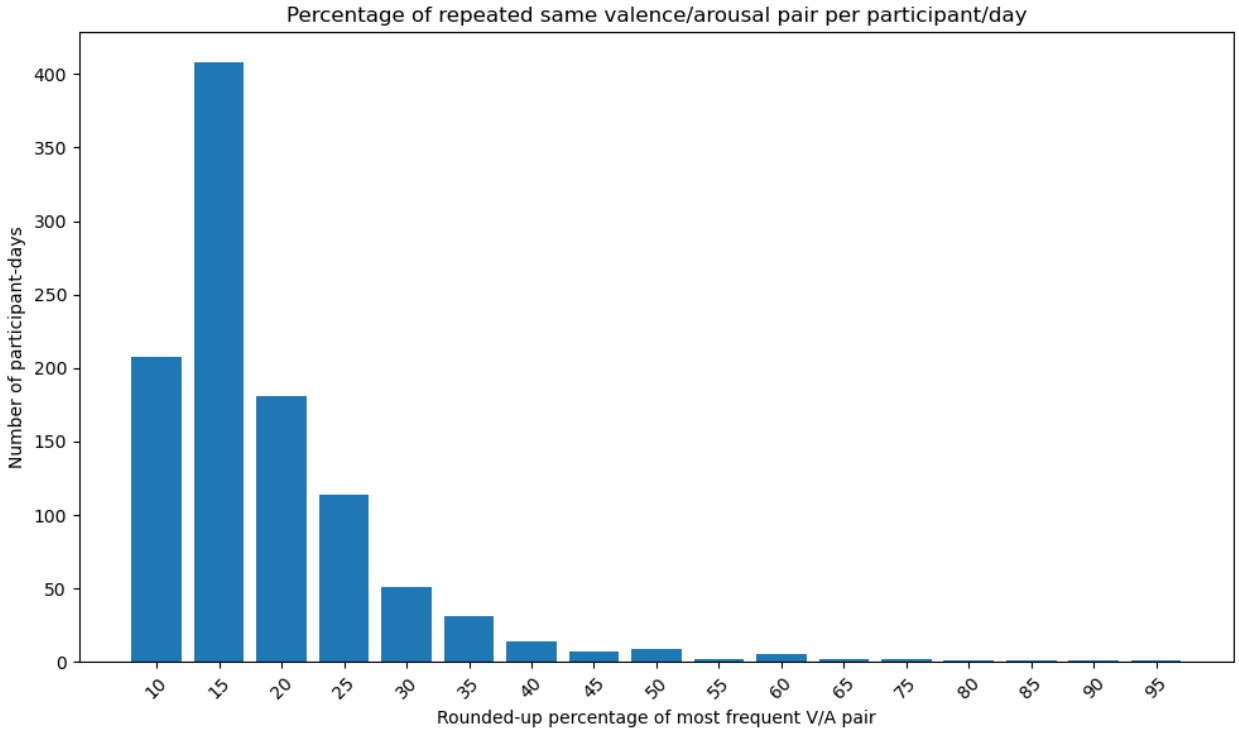
Combined flag criteria

Table 4. Criteria applied when creating the flagged_outlier column.

Criterion	Level / threshold	Filtering consequence
Emotion-label inconsistency	Trial; no compatible label established	Exclude trial
Emotion-level outlier	Trial; Mahalanobis distance > 97th percentile in emotion group	Exclude trial
Video–emotion outlier	Trial; Mahalanobis distance > 97th percentile in video–emotion group	Exclude trial
Rating-dimension repetition	Participant-session; $\|z\| > 1.5$	Exclude session trials
Rating-pair repetition	Participant-session; $\|z\| > 1.5$	Exclude session trials



Supplementary Figure 7. Distribution across participant-days of the percentage of trials with equal rounded valence and arousal ratings.



Supplementary Figure 8. Distribution across participant-days of the percentage of trials with the most frequent rounded valence–arousal pair.